SHRI RAMSWAROOP MEMORIAL COLLEGE OF ENGINEERING AND MANAGEMENT, LUCKNOW

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

HEALTH TRACK - AI INTEGRATED HEALTH DATA AND INFORMATION MANAGEMENT SYSTEM

PROJECT DONE BY : (GROUP - BTP26/AL/02)

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INTRODUCTION

Health Data and Information Management System (**HDIMS**) is a digital platform that centralizes patient health records, appointments, prescriptions, and analytics. It improves accessibility, enhances healthcare decision-making, ensures secure data handling, and enables efficient interaction between patients, doctors, hospitals, and laboratories.

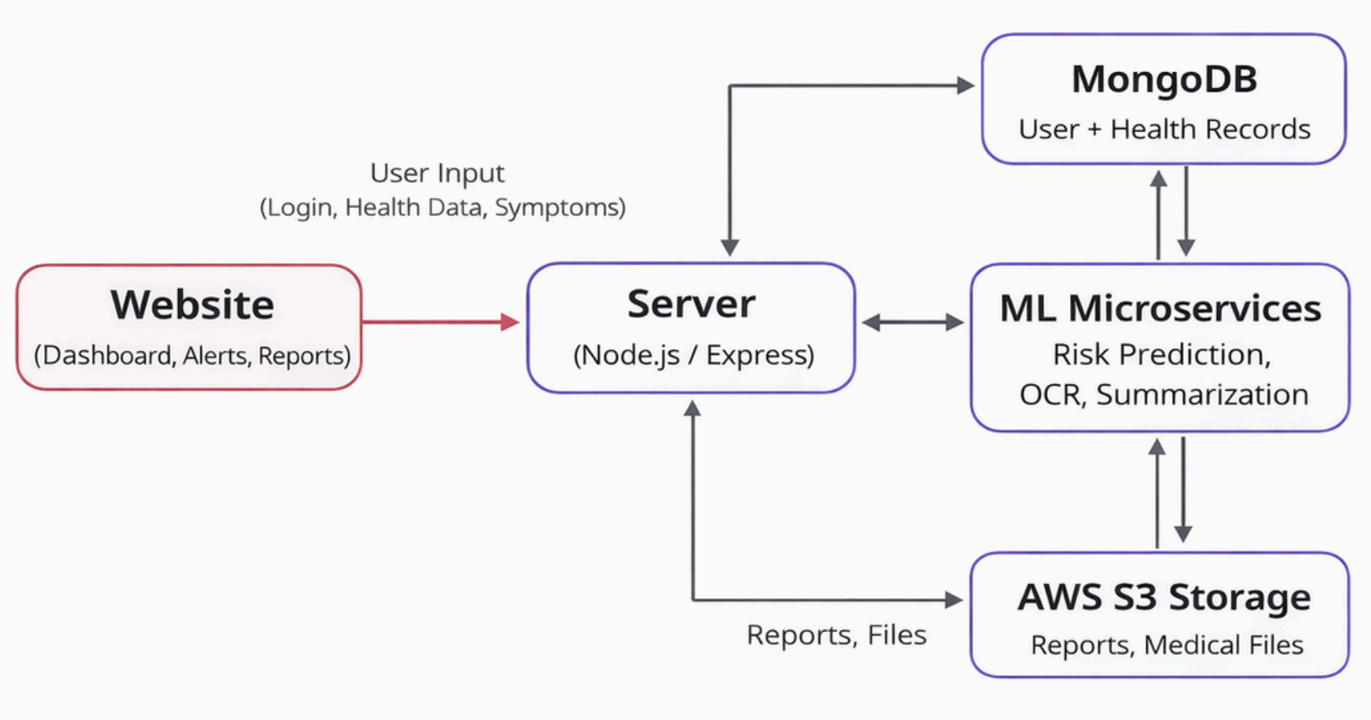
OBJECTIVES

- To develop a centralized system for managing patient health records, including prescriptions and lab reports.
- To provide secure, role-based access for patients, doctors, and administrators.
- To enable easy appointment scheduling and digital prescription management.
- To support AI-based disease prediction and health data analysis through dashboards.

METHODOLOGY

- Agile development approach with iterative updates
- Data collection from users, doctors, and labs
- Centralized database (EHR storage)
- Backend processing using APIs (Node.js/Express)
- AI models for risk prediction and analytics
- Integration of OCR for reports and summarization
- Secure authentication using JWT
- Visualization using dashboards and graphs

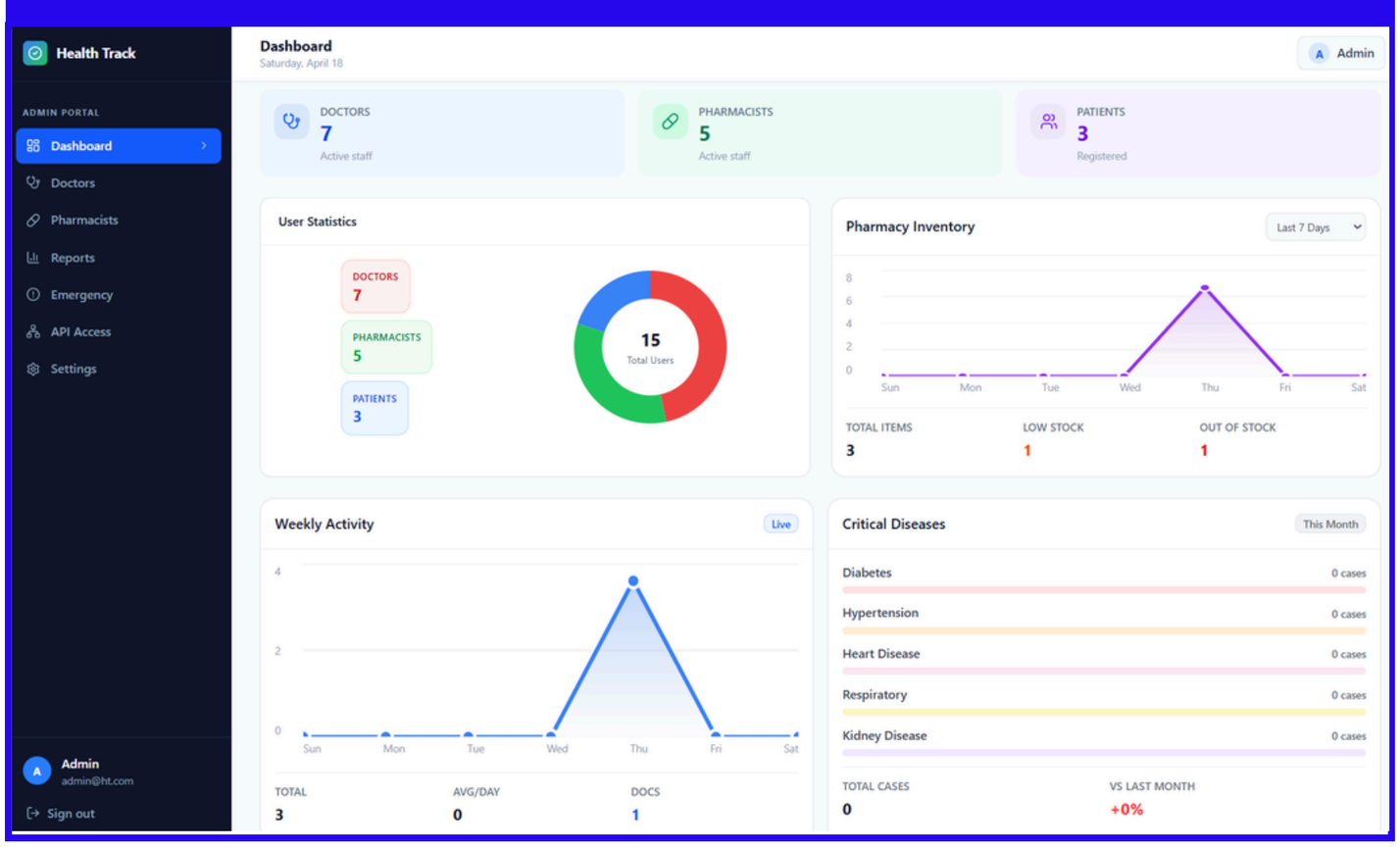
DATA FLOW DIAGRAM



MODULE DESCRIPTION

- **User Management :** Authentication, role-based access (Patient, Doctor, Admin)
- **EHR Module :** Stores prescriptions, reports, allergies, diagnoses
- **Appointment Module :** Booking, scheduling, reminders
- **E-Prescription Module :** Digital prescriptions with verification
- **Lab Integration :** Upload and visualize reports
- **AI Prediction Module :** Disease risk analysis (diabetes, heart disease)
- **Health Tracker :** BMI, vaccination, fitness tracking
- **Analytics Dashboard :** Trends, reports, and insights
- **Security Module :** JWT Based Tokens

SCREENSHOT



ADVANTAGES

- Centralized health data access and visualization
- Encrypted data through AES
- AI-based disease prediction
- Secure and role-based system
- Better patient-doctor communication
- Supports interoperability (HL7/FHIR)

FUTURE SCOPE

- Integration with wearable devices (IoT)
- AI chatbot for health assistance
- Blockchain for secure health records
- Voice assistant for patients
- Disease heatmap visualization
- Personalized healthcare recommendations

CONCLUSION

- Provides a unified healthcare management system
- Enhances decision-making using data and AI
- Improves accessibility and efficiency
- Supports digital transformation in healthcare
- Enables better patient care and monitoring

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